

INSTRUCTION TUNING LARGE LANGUAGE MODELS

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Abstract

Instruction-tuned large language models (LLMs) power many conversational and educational systems but still struggle to reliably follow complex user instructions and to explain technical content at multiple levels of difficulty. This work investigates whether parameter-efficient fine-tuning using LoRA/QLoRA on a blended instruction corpus can enhance the instruction-following capabilities of Mistral-7B while simultaneously adapting it to serve as an AI teaching assistant for the CIS6930 Large Language Models course. The proposed pipeline combines three complementary datasets—UltraChat 200k for multi-turn dialogue, Infinity-Instruct for code- and math-oriented instructions, and Symbolic IT for logic and structured reasoning—and applies a standardized pre-processing and merging strategy before training low-rank adapters on top of a 4-bit quantized Mistral-7B-Instruct backbone.

Experimental evaluation on a 400-example held-out instruction set shows that the tuned model improves mean token-level F1 from roughly 0.31 to 0.34, corresponding to an approximate 10% relative gain, with the largest benefits on moderately difficult Infinity-Instruct tasks. A complementary experiment on CIS6930 lecture slides reports low perplexity and high token-level reconstruction F1, indicating that the model closely captures course material and can reproduce key slide content while retaining fluency. These results suggest that parameter-efficient adaptation with LoRA/QLoRA and a carefully curated blend of instruction datasets is an effective and computationally practical strategy for building course-specific “AI tutors” on top of strong open-source base models, and they motivate future extensions with preference-based training and evaluations on public instruction-following benchmarks.

Introduction

Large language models (LLMs) have rapidly become the backbone of modern conversational agents, coding assistants, and educational tools, yet their usefulness in real-world settings depends critically on how well they can understand and follow user instructions. While recent instruction-tuned models provide strong baselines, they still frequently misinterpret task specifications, omit important constraints, or fail to express domain knowledge at the level of depth and rigor required in technical courses. At the same time, academic environments often lack access to large proprietary instruction datasets and extensive computational resources, making it challenging to reproduce industrial-scale alignment techniques such as full-model fine-tuning and large-scale reinforcement learning from human feedback.

This project addresses these challenges by exploring parameter-efficient instruction tuning of the open-source Mistral-7B-Instruct model using a blended corpus of diverse instruction datasets. The central goal is twofold: to improve the general instruction-following behavior of the model across conversational, code-intensive, and symbolic reasoning tasks, and to specialize it as an AI teaching assistant for the University of Florida's CIS6930 Large Language Models course. To this end, the approach combines UltraChat 200k for multi-turn dialogue, Infinity-Instruct for code and math problems, and Symbolic IT for logic and structured reasoning, followed by an additional specialization stage on lecture slides from the course. Low-rank adaptation (LoRA/QLoRA) is employed to update only a small fraction of the model's parameters on top of a quantized backbone, enabling fine-tuning under constrained GPU memory while preserving the base model's strengths.

Related Work

Method Description

Dataset Processing

Experiment Settings

Experiment Results

Result Analysis

Conclusion